

Hodge and Frobenius colevels of algebraic varieties

Daqing Wan

Dingxin Zhang

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Abstract

We provide new, improved lower bounds for the Hodge and Frobenius colevels of algebraic varieties (over $\mathbf{C}$ or over a finite field) in all cohomological degrees. These bounds are expressed in terms of the dimension of the variety and multi-degrees of its defining equations. Our results lead to an enhanced positive answer to a question raised by Esnault and the first author.

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1 Background and motivation

These notes present our investigation into the cohomological colevels of algebraic varieties in both the Betti and ℓ -adic contexts, motivated by our previous p -adic findings [WZ25]. We provide new, improved lower bounds for the Hodge and Frobenius colevels of algebraic varieties (over $\mathbf{C}$ or over the algebraic closure of a finite field) across all cohomological degrees.

These bounds are formulated in terms of the dimension and the multi-degrees of the defining equations of the variety. For a detailed explanation, refer to Theorems 2.2, 2.3, and 2.5. These theorems offer an enhanced affirmative response to a question (Question 1.11) posed by Esnault and the first author in [EW22].

Before examining the specific results presented in Section 2, this introductory section will review some background material, and illustrate the various aspects of the problems we are about to address.

1.1 Hodge colevel and Frobenius colevel

Let us recall the definitions of *cohomological colevel* in two different contexts.

Definition 1.1 (Frobenius colevel). Let k be an algebraic closure of the prime field $\mathbf{F}_p$, and let $\ell \neq p$ be another prime number. Let X be a finite-type, separated scheme over k . Assume $\mathbf{F}_q$ is a finite subfield of k such that X is defined over $\mathbf{F}_q$, i.e., $X = X_0 \otimes_{\mathbf{F}_q} k$ for some $\mathbf{F}_q$ -scheme X_0 . Let $F_q: X \rightarrow X$ denote the geometric Frobenius (over $\mathbf{F}_q$). The operator F_q acts on the ℓ -adic cohomology $H_c^i(X; \mathbf{Q}_\ell)$ by transporting the structures. We say that $H_c^i(X; \mathbf{Q}_\ell)$ has (Frobenius) *colevel* $\geq m$ if all Frobenius eigenvalues are divisible by q^m in the ring of algebraic integers.

Note that the condition of having a Frobenius colevel $\geq m$ is geometric in nature: if X is defined over $\mathbf{F}_{q'}$, another finite field of characteristic p , then the eigenvalues of F_q acting on $H^i(X; \mathbf{Q}_\ell)$ are divisible by q^m if and only if the eigenvalues of $F_{q'}$ acting on $H^i(X; \mathbf{Q}_\ell)$ are divisible by q'^m .

Definition 1.2 (Hodge colevel). Assume $k = \mathbf{C}$. Let X be a finite-type, separated scheme over $\mathbf{C}$, with associated analytic space X^{an} . The compactly supported cohomology $H_c^i(X^{\text{an}}; \mathbf{Q})$ is endowed with a canonical mixed Hodge structure. We say that $H_c^i(X^{\text{an}}; \mathbf{Q})$ has (Hodge) *colevel* $\geq m$ if the Hodge filtration $F^\bullet H_c^i(X^{\text{an}}; \mathbf{C})$ on $H_c^i(X^{\text{an}}; \mathbf{C})$ satisfies the condition $F^m H_c^i(X^{\text{an}}; \mathbf{C}) = H_c^i(X^{\text{an}}; \mathbf{C})$.¹

In the sequel, we will use the notation $H_c^i(X)$ to represent $H_c^i(X^{\text{an}}; \mathbf{Q})$ or $H_c^i(X; \mathbf{Q}_\ell)$ when the base field k is either irrelevant to the discussion or clearly understood from the context.

If X is proper over k , then the compactly supported cohomology $H_c^*(X)$ is the same as the usual cohomology. In such cases, we will omit the subscript c .

Example 1.3. The colevel of $H^{2i}(\mathbf{P}^n)$ is $\geq i$. If X is a variety over k of dimension $\leq n$, then the colevel of $H_c^{2n}(X)$ is $\geq n$.

When $k = \mathbf{C}$ and X is both smooth and projective, the statement that $H_c^i(X^{\text{an}}; \mathbf{Q}) = H^i(X^{\text{an}}; \mathbf{Q})$ has Hodge colevel $\geq m$ is equivalent to the condition that $H^{i-a}(X; \Omega_X^a) = 0$ for $m > a$. Here, $\Omega_X^a = 0$ if $a < 0$.

1.2 Geometric colevel and cohomological colevel

The cohomological colevel conditions discussed above are connected to the well-known conjectures about the existence of algebraic cycles in the following manner. Let X be a smooth projective variety over k . If there exists a subvariety Z of codimension $\geq m$ such that the map $H_Z^i(X) \rightarrow H^i(X)$ is surjective—where the left-hand side denotes the cohomology of X supported on Z —then we say that the *geometric colevel* of $H^i(X)$ is $\geq m$. For the general definition of the colevel filtration (*la filtration par le coniveau*), the reader is referred to [Gro68, §10.1] or [DK73, Exposé XX, §2]. It can be shown that if the geometric colevel of $H^i(X)$ is $\geq m$, then the cohomological (i.e., Frobenius or Hodge) colevel of $H^i(X)$ is also $\geq m$. See [Gro68, p. 167].

¹These colevel conditions can also be defined for “abstract” mixed Hodge structures or Frobenius modules. Let $r \in \mathbf{Z}$ and K be a field. We say that a filtered K -vector space $(V, \text{Fil}^\bullet V)$ has (Hodge) colevel $\geq r$ if $V = \text{Fil}^r V$. If V is a finite-dimensional $K[F]$ -module, we say that V has (Frobenius) colevel $\geq r$ if all the eigenvalues of F can be expressed as $q^r \cdot \beta$, where β is some algebraic integer.

Dually, we say that a filtered vector space $(V, \text{Fil}^\bullet V)$ has (Hodge) level $\leq r$ if $\text{Fil}^{r+1} V = 0$; we say that a finite-dimensional $K[F]$ -module V has (Frobenius) level $\leq r$ if, for any eigenvalue α of F , $q^r \cdot \alpha^{-1}$ is an algebraic integer.

Example 1.4. Let X be a proper nonsingular variety over k . Let $k(X)$ denote the rational function field of X , and let $\overline{k(X)}$ be an algebraic closure of $k(X)$. Suppose that

$$\mathrm{CH}_0\left(X \times_k \mathrm{Spec} \overline{k(X)}\right) \otimes_{\mathbf{Z}} \mathbf{Q} \simeq \mathbf{Q}$$

(for example, proper nonsingular rationally connected varieties, including all Fano manifolds, satisfy this condition). Then $H^i(X)$ has geometric colevel ≥ 1 for any $i > 0$. Consequently, depending on the context, the Frobenius or Hodge colevel of $H^i(X)$ is also at least 1. For further information, see [Esn03b].

Conversely, if the cohomological colevel of $H^i(X)$ is $\geq m$, the generalized Hodge conjecture in the Betti case, and the generalized Tate conjecture in the ℓ -adic case, state that the geometric colevel is also at least m , see [Gro68, §10] and [Gro69].

Remark 1.5. There is a more sophisticated version of the generalized Hodge conjecture, also proposed by Grothendieck. This version is applicable to both singular and non-proper varieties and is formulated in terms of motives. It states that if M is an “effective mixed motive” whose de Rham realization M_{DR} has colevel at least μ (with respect to the Hodge filtration), then M can be expressed as $N \otimes \mathbf{Q}(-\mu)$, where N is some effective mixed motive and $\mathbf{Q}(-\mu)$ denotes the Tate twist.

It is well known that the existence of a subvariety Z that supports the cohomology (or the existence of N in the more sophisticated conjecture) has significant arithmetic consequences. For instance, when $k = \mathbf{C}$ and X is smooth and projective, the generalized Hodge conjecture suggests that a Hodge colevel of $H^i(X)$ being $\geq m$ implies that a sufficiently general mod p reduction of X should also have Frobenius colevel $\geq m$. Let us elaborate on this for readers who may not be familiar with this circle of ideas.

If $H^i(X)$ has Hodge colevel $\geq m$, the generalized Hodge conjecture predicts the existence of a subvariety Z of X with codimension $\geq m$ such that the following map is surjective:

$$H_{Z^{\mathrm{an}}}^i(X^{\mathrm{an}}; \mathbf{Q}) \rightarrow H^i(X^{\mathrm{an}}; \mathbf{Q}).$$

By the comparison theorem between étale and singular cohomology, the map of ℓ -adic cohomology groups

$$H_Z^i(X; \mathbf{Q}_{\ell}) \rightarrow H^i(X; \mathbf{Q}_{\ell}) \tag{1.6}$$

is also surjective. Now, let us consider a spread-out of X , i.e., a smooth projective morphism $\mathcal{X} \rightarrow \mathrm{Spec} R$ —where R is a finitely generated subring of $\mathbf{C}$ which is smooth over $\mathbf{Z}$ —such that $\mathcal{X} \otimes_R \mathbf{C} = X$. Note that the residue fields of the maximal ideals of R are finite fields.

Upon replacing $\mathrm{Spec} R$ by a smaller nonempty open affine, we may assume the Zariski closure $\mathcal{Z}$ of Z is flat over R . For any closed point b of $\mathrm{Spec} R$, we form the following cartesian diagram

$$\begin{array}{ccccc} Z & \longrightarrow & \mathcal{Z} & \longleftarrow & Z_b \\ \downarrow & & \downarrow & & \downarrow \\ X & \longrightarrow & \mathcal{X} & \longleftarrow & X_b \\ \downarrow & & \downarrow & & \downarrow \\ \mathrm{Spec} \mathbf{C} & \longrightarrow & \mathrm{Spec} R & \longleftarrow & \{b\} \end{array} .$$

We shall refer to X_b as a *specialization* of X . Let $\bar{b}$ denote a geometric point of $\mathrm{Spec}(R)$ valued in an algebraic closure of the residue field $\kappa(b)$ of the closed point b .

By generic base change [Del77, Finitude], there exists a Zariski open dense subscheme of $\mathrm{Spec} R$ over which the formation of $H_{Z_b}^i(X_b, \mathbf{Q}_\ell)$ commutes with arbitrary base change. Therefore, for b lying in this open set, we have isomorphisms of vector spaces:

$$H_Z^i(X; \mathbf{Q}_\ell) \simeq H_{Z_b}^i(X_b; \mathbf{Q}_\ell),$$

and

$$H^i(X; \mathbf{Q}_\ell) \simeq H^i(X_b; \mathbf{Q}_\ell).$$

Therefore, the surjectivity of (1.6) implies that, for a “sufficiently general” closed point b of $\mathrm{Spec} R$, the map

$$H_{Z_b}^i(X_b; \mathbf{Q}_\ell) \rightarrow H^i(X_b; \mathbf{Q}_\ell)$$

is surjective as well. Thus, $H^i(X_b; \mathbf{Q}_\ell)$ has geometric colevel $\geq m$; and it follows that $H^i(X_b; \mathbf{Q}_\ell)$ has Frobenius colevel $\geq m$, i.e., the Frobenius eigenvalues of X_b are divisible by q^m , where q is the cardinality of $\kappa(b)$.

In particular, if $H^i(X; \mathcal{O}_X) = 0$ for all $i > 0$, and if we assume the generalized Hodge conjecture is valid, then a sufficiently general specialization X_b should have a rational point according to Grothendieck’s trace formula. Furthermore, the number of $\kappa(b)$ -points is congruent to 1 modulo q , where $q = |\kappa(b)|$.

This prediction for smooth projective X over $\mathbf{C}$, based on the generalized Hodge conjecture, aligns with a more refined conjecture—“Newton above Hodge”—formulated by Katz [Kat71, Conjecture 2.9]. This latter conjecture has been unconditionally proven by Mazur [Maz72].

Remark 1.7. One cannot expect the Hodge colevel of $H^i(X)$ to be *equal* to the Frobenius colevel of $H^i(X_b)$ for a sufficiently general smooth specialization X_b . A well-known example illustrates this: According to a theorem by Elkies [Elk87], any elliptic curve over $\mathbf{Q}$ has infinitely many supersingular reductions. Therefore, for any elliptic curve E over $\mathbf{Z}[1/N]$, there are infinitely many primes p such that $H^2(E_{\overline{\mathbf{F}}_p} \times E_{\overline{\mathbf{F}}_p})$ has Frobenius colevel 1, while the Hodge colevel of $H^2(E_{\mathbf{C}} \times E_{\mathbf{C}})$ is zero.

On the other hand, one does expect that the Hodge colevel of $H^i(X)$ should be equal to the Frobenius colevel of $H^i(X_b)$ for *infinitely* many specializations X_b (more generally, it is expected that X should have infinitely many “ordinary” reductions). The case of colevel zero corresponds to the weak ordinarity conjecture proposed by Mustařa and Srinivas [MS11]. Proving this in general appears to be quite challenging. According to the generalized Tate conjecture, this expectation is related to the potential equality between the geometric colevel of $H^i(X)$ and the geometric colevel of $H^i(X_b)$ for infinitely many specializations X_b .

When X_b is a *singular* specialization, the colevel bounds on X still have important arithmetic implications. We refer the reader to Esnault [Esn06] and Berthelot–Esnault–Rülling [BER12] for investigations around this theme.

1.3 Common lower bounds for cohomological Colevels

More generally, for *any* (without smooth proper condition) flat morphism $\mathcal{X} \rightarrow \mathrm{Spec} R$, with $R \subset \mathbf{C}$ finitely generated over $\mathbf{Z}$, and for b being “sufficiently general”, the Hodge Colevels of X are not larger than the Frobenius Colevels of X_b^2 .

²This can be seen, for instance, by spreading out a proper hypercovering of the generic fiber of $\mathcal{X} \rightarrow \mathrm{Spec} R$, whose entries are normal crossing complements of smooth projective varieties.

This conceptual explanation, appealing as it is, is not very practical. For instance, it cannot even be used to deduce the classical theorem of Ax and Katz [Ax64, Kat71] (on Frobenius colevels of primitive cohomology of a *possibly singular* projective variety) from a Hodge-theoretic result (cf. [ENS92]): we could indeed lift the defining equations, but we have no idea whether the prime we start with is “sufficiently general”.

Therefore, it is desirable to establish *common* lower bounds for the Hodge colevel of $H_c^i(X)$ and the Frobenius colevel of $H_c^i(X_b)$, regardless of whether b is general or not, and irrespective of how singular the specialization X_b may be. Ideally, this should be done in a purely formal manner, so that the argument is applicable in both the Frobenius and Hodge contexts. Such lower bounds have been extensively studied in the literature, with a notable example being the following integrality theorem of Deligne.

Theorem 1.8 (Deligne). *Let k be either a finite field, or $\mathbf{C}$. Let X be a finite type, separated scheme over k of dimension n . Then for any integer $j \in \mathbf{Z}$, $H_c^{n+j}(X)$ has colevel $\geq \max\{0, j\}$.³*

The bounds provided by Deligne’s theorem are quite crude, as they apply to *any* variety over k . To improve upon this, one could consider the type of polynomial equations that define X and investigate whether the colevels can be bounded from below by quantities that depend solely on these equations. One set of such quantities consists of the following numbers. These numbers appear, for instance, in the classical Ax–Katz theorem [Ax64, Kat71] and in the first author’s work [Wan00] on the “polar part” of zeta functions of algebraic varieties.

Definition 1.9. For a sequence of integers $d_1 \geq \dots \geq d_r \geq 1$ and integers $N, j \geq 0$, define

$$\mu_j(N; d_1, \dots, d_r) = j + \max \left\{ 0, \left\lfloor \frac{N - j - \sum_{i=1}^r d_i}{d_1} \right\rfloor \right\}.$$

Note that we have $\mu_j(N; d_1, \dots, d_r) = j + \mu_0(N - j; d_1, \dots, d_r) \geq j$, and the sequence $\mu_j(N; d_1, \dots, d_r)$ is increasing in j :

$$\mu_0(N; d_1, \dots, d_r) \leq \mu_1(N; d_1, \dots, d_r) \leq \mu_2(N; d_1, \dots, d_r) \leq \dots$$

The following theorem summarizes the best known lower bounds in this regard, which are the result of the contribution from several mathematicians, including Ax, Katz, Deligne, Dimca, Esnault, Nori, Srinivas, del Angel, the first author, among others.

(For a closed subvariety Y of $\mathbf{P}^N$, set $H^*(Y)_{\text{prim}} = \text{Coker}(H^*(\mathbf{P}^N) \rightarrow H^*(Y))$.)

Theorem 1.10. *Let k be either an algebraic closure of a finite field, or the field $\mathbf{C}$ of complex numbers.*

However, one cannot expect this to hold for every b , even when the generic fiber of $\mathcal{X} \rightarrow \text{Spec } R$ is smooth and projective. For a counterexample, see [EX09, Theorem 1.2].

³The finite field case was proved by Deligne in [DK73, Exposé XXI, Corollaire 5.5.3]. The theorem for Hodge colevels (which is not needed in this paper) can be shown as follows. When X is nonsingular, the theorem follows from Poincaré duality and Deligne’s *definition* of Hodge filtration in terms of logarithmic forms; if X is possibly singular, one proceeds by induction on dimension, and use the long exact sequence

$$\dots \rightarrow H_c^{n+j}(X - X_{\text{sing}}) \rightarrow H_c^{n+j}(X) \rightarrow H_c^{n+j}(X_{\text{sing}}) \rightarrow \dots,$$

where X_{sing} is the singular locus of the reduced structure of X , which has lower dimension. Applying the theorem for nonsingular variety to $X - X_{\text{sing}}$, and inductive hypothesis to X_{sing} finishes the proof.

1. (Affine Case). Let X be the affine variety in $\mathbf{A}^N$ of dimension n defined by $f_1 = \cdots = f_r = 0$, where $f_i \in k[x_1, \dots, x_n]$, has degree $\leq d_i$.

(a) The colevels of $H_c^*(X)$ and $H_c^*(\mathbf{A}^N - X)$ are at least $\mu_0(N; d_1, \dots, d_r)$, and

(b) for any $0 \leq j \leq N - 1$, the colevels of

$$H_c^{N-1+j}(X) \quad \text{and} \quad H_c^{N+j}(\mathbf{A}^N - X)$$

are at least $\mu_j(N; d_1, \dots, d_r)$.

2. (Projective Case). Let Y be the projective variety in $\mathbf{P}^N$ of dimension n defined by $g_1 = \cdots = g_r = 0$, where $g_i \in k[x_0, \dots, x_N]$ is homogeneous of degree $\leq d_i$.

(a) The colevels of $H^*(Y)_{\text{prim}}$ and $H_c^*(\mathbf{P}^N - Y)$ are at least $\mu_0(N + 1; d_1, \dots, d_r)$, and

(b) for $0 \leq j \leq n$, the colevels of $H^{n+j}(Y)_{\text{prim}}$ and $H_c^{n+1+j}(\mathbf{P}^N - Y)$ are at least $\mu_j(N + 1; d_1, \dots, d_r)$.

1.4 A brief history of Theorem 1.10

Item (1a) for Frobenius colevel was inspired by the classical theorem of Ax and Katz [Ax64, Kat71], which asserts that the reciprocal zeros and poles of the zeta function $Z_X(t)$ of X are divisible by $\mu_0(N; d_1, \dots, d_r)$ in the ring of algebraic integers. But the Ax–Katz theorem does not directly imply (1a), as there may be cancellation of Frobenius eigenvalues in the zeta function. The form as stated is due to Esnault and Katz [EK05, Theorems 2.1–2.3 and §5.3], based on a method developed by del Angel [dAR94]. This lower bound is essentially optimal in general, as the Ax–Katz bound is sharp.

Item (2a) for Hodge colevels was proved by Deligne in [DK73, Exposé XI] when Y is a smooth complete intersection, and historically this result motivated Katz’s improvement of Ax’s theorem. For an arbitrary hypersurface Y , (2a) for Hodge colevel was proved by Deligne and Dimca [DD90]. For an arbitrary complete intersection Y , (2a) for Hodge colevel was proved by Esnault [Esn90]. For an arbitrary projective Y , (2a) for Hodge colevel was proved by Esnault–Nori–Srinivas [ENS92].

For Frobenius colevels, Esnault [Esn03a] proved (2a) for complete intersections. The general result (2a) is proved by Esnault and Katz [EK05, Theorem 4.1(1)]. This part of the estimate is essentially sharp in general, again by Ax–Katz and by Deligne’s result for smooth complete intersections above.

As hinted by Deligne’s Theorem 1.8, compactly supported cohomology groups beyond the middle dimension tend to have larger colevels. Thus, Items (1b) and (2b) both reflect this phenomenon.

In a different related direction, the first author [Wan00] made some early attempts in the framework of zeta functions via Dwork’s p -adic theory. He showed that, in the affine case, the reciprocal poles of $Z_X(t)^{(-1)^{\dim X+1}}$ are divisible by $q^{\mu_1(N; d_1, \dots, d_r)}$ as algebraic integers, when X is a complete intersection. This improved the “polar part” of the Ax–Katz theorem for complete intersections.

Motivated by this, (2b) for Hodge colevel was shown by Esnault and the first author [EW03] for projective complete intersection Y . Esnault–Katz [EK05] proved, for an arbitrary projective Y , that the colevel of $H_c^{N+j}(\mathbf{P}^N - Y)$ is $\geq \mu_j(N + 1; d_1, \dots, d_r)$. This is more general but weaker than (2b), as the cohomological degree of the improved bound starts from the bigger middle dimension N of

$\mathbf{P}^N - Y$, not the smaller number $n + 1$. The sharper form (2b) in the general case was proved by Esnault and the first author [EW22].

1.5 Better colevel bounds for affine varieties

Similarly, Item (1b) is not an optimal estimate either, unless X is a hypersurface, as the cohomological degree of the improved bound starts from the bigger number $N - 1$, not the smaller middle dimension n of X .

In [EW22], the following improvement of (1b) was proposed as an open question. It is the affine analogue of the better bound (2b) for projective varieties:

Question 1.11 ([EW22]). Under the hypothesis of Theorem 1.10(1), is it true that for $0 \leq j \leq n$, the colevels of $H_c^{n+j}(X)$ and $H_c^{n+j+1}(\mathbf{A}^N - X)$ are at least $\mu_j(N; d_1, \dots, d_r)$?

In [EW22], Esnault and the first author gave an evidence that this question should have a positive answer: if the defining equations of X have a certain good behavior at infinity, then one can reduce the bounds proposed in the question to Theorem 1.10(2b). It is also noted that if a suitable version of weak Lefschetz theorem holds, then the Question has an affirmative answer.

In these notes, we prove the needed Lefschetz type theorem. See Lemma 3.9. As a consequence, we get the following⁴.

Theorem 1.12 (See Lemma 4.2). *Question 1.11 has an affirmative answer. That is, for integer $0 \leq j \leq n$, the colevels of $H_c^{n+j}(X)$ and $H_c^{n+j+1}(\mathbf{A}^N - X)$ are at least $\mu_j(N; d_1, \dots, d_r)$.*

Since $\mu_j(N; d_1, \dots, d_r) \geq j$, Theorem 1.12 improves Deligne’s integrality theorem (Theorem 1.8) for affine variety X . Also, this theorem gives a cohomological interpretation of the polar result in [Wan00] on zeta functions of complete intersections.

While Theorem 1.12 confirms Question 1.11, the bounds are still not optimal. Moreover, it does not provide information of colevels of $H_c^*(X)$ before the middle cohomological dimension if X is not a complete intersection. We shall establish new lower bounds of the colevels of $H_c^i(X)$ for *all* i . See Theorems 2.2 and 2.3. For the cohomology groups beyond middle dimension, our new bounds will be sharper than the ones proposed in the question above and hence better than both Theorem 1.10(1b) and Theorem 1.12; for cohomology groups before middle dimension, our lower bounds will improve the Ax–Katz type bound of Theorem 1.10(1a). Examples 6.1 and 6.2 in the final section give concrete illustrations of how our new bounds are improving the bounds provided by Theorem 1.10 and Theorem 1.12.

From our refined colevel bounds for affine varieties, one deduces formally “projective bounds” improving Theorem 1.10(2a, 2b). See Theorem 2.5.

These refined lower bounds were first discovered in our previous Dwork-theoretic study of Frobenius eigenvalues of rigid cohomology [WZ25]. We noticed there that our p -adic method is not “formal” enough to provide colevel bounds for ℓ -adic and Betti cohomology.

The proof we present in these notes combines several ideas: the “Gysin lemma” (Lemma 3.9) already mentioned; a dévissage designed by del Angel [dAR94] and Esnault–Katz [EK05] which allows us to reduce the number of defining equations without jeopardizing the lower bound, see Lemma 4.7; and

⁴After we finish the manuscript, we learned that Rai and Shuddhodan also obtained a proof of Theorem 1.12 [RS23].

finally a crucial algebraic lemma devised in [WZ25] which shows that the “core” of the improved lower bounds comes from a set-theoretic complete intersection, see Lemma 5.2.

2 Improved colevel bounds

To state our refined colevel bounds, let us introduce some sequences of numbers.

Definition 2.1. Let $d_1, d_2, \dots, d_r$ and N be positive integers. Assume that $d_1 \geq d_2 \geq \dots \geq d_r$. For each integer $e \in \mathbf{Z}$, and $i = 1, 2, \dots, r$, we define

$$d_i^*(e) = \begin{cases} d_i, & \text{if } i \leq e, \\ 1, & \text{if } i > e, \text{ and } d_i = d_1, \\ 0, & \text{if } i > e, \text{ and } d_i < d_1. \end{cases}$$

So if $e \leq 0$, $d_i^*(e) = 1$ or 0 depending whether $d_i = d_1$ or $d_i < d_1$. If $e \geq r$, then $d_i^*(e) = d_i$.

Define

$$v_j^{(e)}(N; d_1, \dots, d_r) = j + \max \left\{ 0, \left\lfloor \frac{N - j - \sum_{i=1}^r d_i^*(e)}{d_1} \right\rfloor \right\}.$$

Since $d_i^*(e)$ is increasing in e , it is clear that $v_j^{(e)}(N; d_1, \dots, d_r)$ is decreasing in e . This gives the decreasing relation

$$\begin{aligned} \dots &= v^{(-1)}(N; d_1, \dots, d_r) = v^{(0)}(N; d_1, \dots, d_r) \\ &\geq v_j^{(1)}(N; d_1, \dots, d_r) \geq \dots \geq v_j^{(r)}(N; d_1, \dots, d_r) = v_j^{(r+1)}(N; d_1, \dots, d_r) = \dots \\ &= \mu_j(N; d_1, \dots, d_r). \end{aligned}$$

Our improved colevel bounds beyond middle dimension is the following.

Theorem 2.2. Let X be an n -dimensional Zariski closed subset of $\mathbf{A}^N$ defined by polynomials $f_1 = \dots = f_r = 0$. Let $d_1 \geq \dots \geq d_r$ be a sequence of positive integers such that $d_i \geq \deg f_i$. Then the colevels of $H_c^{n+j}(X)$ and $H_c^{n+j+1}(\mathbf{A}^N - X)$ are at least $v_j^{(N-n)}(N; d_1, \dots, d_r)$ for all $0 \leq j \leq n$.

Our improved colevel bounds before middle dimension is the following.

Theorem 2.3. In the situation of Theorem 2.2, for $0 \leq m \leq r - (N - n)$, the colevels of $H_c^{N-r+m}(X)$ and $H_c^{N-r+m+1}(\mathbf{A}^N - X)$ are at least $v_0^{(r-m)}(N; d_1, \dots, d_r)$.

Under the hypothesis of Theorems 2.2 and 2.3, we have $H_c^i(X) = 0$ for $i < N - r$ (this is well-known, an elementary argument is given in [WZ25, Lemma 6.2]); also by general facts about cohomology, we have $H_c^{n+j}(X) = 0$ for $j > n$. Thus the two theorems cover all possibly nonzero compactly supported cohomology groups of X . The integer $r - (N - n)$ is always non-negative. It is zero if and only if X is a set-theoretic complete intersection by r equations.

The assertions for the complement $\mathbf{A}^N - X$ follow from the results about X by using the long exact sequence

$$\dots \rightarrow H_c^i(\mathbf{A}^N - X) \rightarrow H_c^i(\mathbf{A}^N) \rightarrow H_c^i(X) \rightarrow \dots$$

and the well-known fact that

$$H_c^i(\mathbf{A}^N) = \begin{cases} 0, & i \neq 2N, \\ \Lambda(-N), & i = 2N. \end{cases}$$

Thus, Theorems 2.2 and 2.3 cover all possibly nonzero compactly supported cohomology of $\mathbf{A}^N - X$ except the top-dimensional ($i = 2N$) one, which is trivially of colevel N .

Since $\nu_j^{(N-n)}(N; d_1, \dots, d_r) \geq \mu_j(N; d_1, \dots, d_r)$, we see Theorem 2.2 implies, under its hypothesis, that

Corollary 2.4. *The colevel of $H_c^{n+j}(X)$ is at least $\mu_j(N; d_1, \dots, d_r)$.*

Thus, Theorem 2.2 gives an affirmative answer to Question 1.11 and hence also improves Theorem 1.12. For if X is a complete intersection by given r equations, namely, $r = N - n$, one has $\nu_j^{(N-n)}(N; d_1, \dots, d_r) = \mu_j(N; d_1, \dots, d_r)$. In this case, Theorem 2.2 is the same as Theorem 1.12. Thus, strict improvements of Theorem 2.2 over Theorem 1.12 can only occur for non-complete intersections. See Examples 6.1 and 6.2.

Similarly, since $\nu_0^{(r-m)}(N; d_1, \dots, d_r) \geq \mu_0(N; d_1, \dots, d_r)$, the lower bounds given by Theorem 2.3 are improvements of the Ax-Katz type bound $\mu_0(N; d_1, \dots, d_r)$.

From Theorems 2.2 and 2.3, we can formally deduce the following projective version which is an improvement of Theorem 1.10(2a, 2b).

Theorem 2.5. *Let Y be a Zariski closed subset of $\mathbf{P}^N$ of dimension n , defined by homogeneous polynomial equations $g_1 = \dots = g_r = 0$. Assume that $\deg g_i \leq d_i$, $d_1 \geq \dots \geq d_r$. Then*

- *for all integers $0 \leq j \leq n$, the colevels of $H^{n+j}(Y)_{\text{prim}}$ and $H_c^{n+j+1}(\mathbf{P}^N - Y)$ are at least $\nu_j^{(N-n)}(N + 1; d_1, \dots, d_r)$;*
- *for all integers $0 \leq m \leq r - (N - n)$, the colevels of $H^{N-r+m}(Y)_{\text{prim}}$ and $H_c^{N-r+m+1}(\mathbf{P}^N - Y)$ are at least $\nu_0^{(r-m)}(N + 1; d_1, \dots, d_r)$.*

It is worth remarking that we *do not* give new proofs of Theorems 1.10(1a, 2a) (which are ultimately based on the Ax-Katz theorem, a theorem whose known proofs are all p -adic in nature). Rather, these results are the starting point for our proofs of Theorems 2.2 and 2.3.

As an arithmetic application, we give the following corollary which improves the “polar” part of the Ax-Katz theorem for the zeta function of an arbitrary affine variety, extending the complete intersection case treated in [Wan00]. This result can also be derived from the corresponding Frobenius colevel bounds in rigid cohomology [WZ25].

Corollary 2.6. *In the situation of Theorem 2.2, assume now that k is a finite field of q elements.*

1. *If $r = N - n$ (complete intersection), then all reciprocal poles of the zeta function $Z_X(t)^{(-1)^{N-r+1}}$ are divisible, as algebraic integers, by $q^{\mu_1(N; d_1, \dots, d_r)}$.*
2. *If $r > N - n$ (non-complete intersection), then all reciprocal poles of the zeta function $Z_X(t)^{(-1)^{N-r+1}}$ are divisible, as algebraic integers, by $q^{\nu_0^{(r-1)}(N; d_1, \dots, d_r)}$.*

Proof. The following sequence of colevel bounds is increasing:

$$\nu_0^{(r)}(N; d_1, \dots, d_r) \leq \dots \leq \nu_0^{(N-n)}(N; d_1, \dots, d_r) \leq \nu_1^{(N-n)}(N; d_1, \dots, d_r) \leq \dots$$

By the trace formula, the zeta function $Z_X(t)^{(-1)^{N-r+1}}$ is given by the product

$$\prod_{m=0}^{r-(N-n)} \det(I - t \cdot F \mid H_c^{N-r+m}(X))^{(-1)^m} \prod_{j=1}^n \det(I - t \cdot F \mid H_c^{n+j}(X))^{(-1)^{N-r+n+j}},$$

where F is the geometric Frobenius map. The factor for $m = 0$ is clearly in the numerator. By Theorems 2.2 and 2.3, the reciprocal roots of all other factors (including all polar factors) are divisible, as algebraic integers, by $q^{\nu_0^{(r-1)}(N; d_1, \dots, d_r)}$ ($m = 1$) if $r > N - n$, and by $q^{\nu_1^{(N-n)}(N; d_1, \dots, d_r)}$, which is $q^{\mu_1(N; d_1, \dots, d_r)}$, if $r = N - n$. ■

We close this section by proving the following very elementary lemma, which summarizes some basic properties of the numbers $\nu_j^{(e)}(N; d_1, \dots, d_r)$. These properties will be used in our proof.

Lemma 2.7. *The numbers $\nu_j^{(e)}(N; d_1, \dots, d_r)$ satisfy the following properties.*

1. $\nu_j^{(r)}(N; d_1, \dots, d_r) = \mu_j(N; d_1, \dots, d_r)$.
2. $\nu_j^{(e)}(N; d_1, \dots, d_r)$ is decreasing in e and d_i , increasing in j .
3. For any $e \in \mathbf{Z}$, $i \geq 1$, $j \geq 0$, we have

$$\nu_{j+i}^{(e)}(N; d_1, \dots, d_r) \geq \nu_j^{(e-i)}(N; d_1, \dots, d_r).$$

4. Suppose $1 \leq e \leq r_1 \leq r - 1$, then

$$\nu_{j+(r-r_1-1)}^{(e)}(N; d_1, \dots, d_{r_1}) \geq \nu_j^{(e)}(N; d_1, \dots, d_r).$$

Proof. The proofs of (1) and (2) are straightforward and omitted. Clearly, it suffices to prove (3) for $i = 1$. We first assume $1 \leq e \leq r$. In this case, we have following estimate:

$$\begin{aligned} \nu_{j+1}^{(e)}(N; d_1, \dots, d_r) &= j + 1 + \max \left\{ 0, \left\lceil \frac{N - j - 1 - \sum_i d_i^*(e)}{d_1} \right\rceil \right\} \\ &= j + \max \left\{ 1, \left\lceil \frac{N - j + (d_1 - d_e) - \sum_{i \neq e} d_i^*(e) - 1}{d_1} \right\rceil \right\} \\ &\geq j + \max \left\{ 1, \left\lceil \frac{N - j - \sum_i d_i^*(e - 1)}{d_1} \right\rceil \right\} \\ &\geq \nu_j^{(e-1)}(N; d_1, \dots, d_r). \end{aligned}$$

If $e \geq r + 1$, then $d_i(e) = d_i = d_i(e - 1)$, the property reduces to the inequality

$$\max \left\{ 0, \frac{N - j - 1 - \sum_1^r d_\nu}{d_1} \right\} + j + 1 \geq j + \max \left\{ 0, \frac{N - j - \sum_1^r d_\nu}{d_1} \right\},$$

which is clearly valid since $d_1 \geq 1$. If $e \leq 0$, then the property reduces to

$$\max \left\{ 0, \frac{N - j - 1 - M \cdot d_1}{d_1} \right\} + j + 1 \geq \max \left\{ 0, \frac{N - j - M \cdot d_1}{d_1} \right\} + j$$

where $M = \text{Card}\{\nu : d_\nu = d_1\}$. This is also valid since $d_1 \geq 1$.

To prove (4), we simply compute:

$$\begin{aligned} &\nu_{j+(r-r_1-1)}^{(e)}(N; d_1, \dots, d_{r_1}) \\ &= j + (r - r_1 - 1) + \max \left\{ 0, \left\lceil \frac{N - j - (r - r_1 - 1) - \sum_{i=1}^{r_1} d_i^*(e)}{d_1} \right\rceil \right\} \\ &\geq j + \max \left\{ 0, \left\lceil \frac{N - j + (r - r_1 - 1)(d_1 - 1) + \sum_{i=r_1+1}^r d_i^*(e) - \sum_{i=1}^{r_1} d_i^*(e)}{d_1} \right\rceil \right\} \\ &\geq \nu_j^{(e)}(N; d_1, \dots, d_r) \end{aligned}$$

The same proof shows that for integer $i \geq 0$ and $1 \leq e \leq r_1 \leq r - i$, then

$$v_{j+(r-r_1-i)}^{(e)}(N; d_1, \dots, d_{r_1}) \geq v_j^{(e)}(N; d_1, \dots, d_r).$$

But we will only need the case $i = 1$ as stated in the lemma. This completes the proof. $\blacksquare$

Conventions

In the following, k will either be an algebraic closure finite field, or the field $\mathbf{C}$ of complex numbers. We fix a prime ℓ different from the characteristic of k .

If k is an algebraic closure of a finite field, and if K is a constructible $\mathbf{Q}_\ell$ -complex on a variety X (“arithmetic situation”), we will assume that K is equipped with a Frobenius action (a complex of “Weil sheaves”). This is the case when $X = X_0 \otimes_{\mathbf{F}_q} k$ is defined over some finite subfield $\mathbf{F}_q$, and K is the inverse image of some $\mathbf{Q}_\ell$ -complex on X_0 .

If $k = \mathbf{C}$ (“Betti situation”), and if K is a constructible complex with $\mathbf{Q}$ -coefficient on X^{an} , then we use the symbol $H_c^*(X; K)$ and $H^*(X; K)$ to denote the cohomology of the complex K on the analytic space X^{an} , with or without compact supports. We will assume that K underlies a mixed Hodge module, so that $H_c^*(X; K)$ and $H^*(X; K)$ are a mixed Hodge structures.

In either case, the constant sheaf $\mathbf{Q}_\ell$ or $\mathbf{Q}$ will be denote by Λ , and $\Lambda(m)$ denotes the m^{th} Tate twist. When we write $H_{(c)}^i(X)$, we mean $H_{(c)}^i(X; \Lambda)$.

We shall construct and use several maps between cohomology groups in the main text. They all come from the standard 6-operations. Therefore, in the arithmetic situation, these maps are easily shown to be compatible with the Frobenius actions, once we choose a field of definition of the variety and the complexes; in the Betti situation, these maps are maps of Hodge structures.

3 Generic base change, weak Lefschetz, and a Gysin lemma

The purpose of this section is to deduce, from Deligne’s generic base change theorem 1.8 and weak Lefschetz theorem 3.6, a “Gysin lemma” 3.9. This lemma will be the foundation of the remainder of this paper.

First, recall Deligne’s “generic base change theorem” ([Del77, Finitude, Corollary 2.9]).

Theorem 3.1 (Deligne). *Let S be a finite type, separated scheme over k . Let $f: X \rightarrow Y$ be a morphism of finite type, separated S -schemes. Let K be a constructible complex on either Y or X . Then there exists a dense open subscheme U of S , such that Rf_*K , $Rf_!K$, $f^{-1}K$, $Rf^!K$ are all constructible complexes on X or Y , and the formation is compatible with all base change $S' \rightarrow U \subset S$.*

From the generic base change theorem we deduce the following “generic purity lemma”.

Lemma 3.2. *Let $f: X \rightarrow \mathbf{P}^N$ be a quasi-finite morphism. Let K be a constructible complex of Λ -modules on X . For a hyperplane B , form the fiber diagram*

$$\begin{array}{ccc} X \times_{\mathbf{P}^N} B & \xrightarrow{\iota'_B} & X \\ \downarrow f' & & \downarrow f \\ B & \xrightarrow{\iota_B} & \mathbf{P}^N \end{array} .$$

If B is sufficiently general, then the purity morphism $K|_{X \times_{\mathbf{P}^N} B}[-2](-1) \xrightarrow{\sim} R(\iota'_B)^!K$ is an isomorphism.

Proof. First, assume that f is a finite morphism. Let $\mathcal{M}$ be the space of hyperplanes in $\mathbf{P}^N$ (i.e., $\mathcal{M}$ is the dual projective space $\check{\mathbf{P}}^N$). Form the incidence correspondence $\mathcal{A} = \{(x, [B]) \in \mathbf{P}^N \times \mathcal{M} : x \in B\}$. Let $\iota: \mathcal{A} \rightarrow \mathbf{P}_{\mathcal{M}}^N = \mathbf{P}^N \times \mathcal{M}$ be the closed immersion. Then the projections $\text{pr}_1: \mathbf{P}_{\mathcal{M}}^N \rightarrow \mathbf{P}^N$, and $\text{pr}_1 \circ \iota: \mathcal{A} \rightarrow \mathbf{P}^N$ are all smooth, equidimensional morphisms. Let $L = \text{pr}_1^{-1}(Rf_*K)$. Since L is the inverse image of a complex by a smooth, equidimensional morphism, and since $\text{pr}_1 \circ \iota$ is also smooth and equidimensional, the relative purity theorem implies that the purity morphism is an isomorphism

$$\iota^{-1}L \xrightarrow{\sim} R\iota^!L[2](1). \quad (3.3)$$

See, for example, [KW01, Theorem 11.2, Supplement].

By Theorem 3.1, there is a Zariski dense open subset $\mathcal{U}$ of $\mathcal{M}$ such that the formation of $R\iota^!L$ commutes with arbitrary base change $S \rightarrow \mathcal{U}$. In particular, for any $[B] \in \mathcal{U}$, we have

$$(R\iota^!L)|_B \simeq R\iota_B^!(L|_{\mathbf{P}^N \times \{[B]\}}). \quad (3.4)$$

Here we are using the following base change diagram

$$\begin{array}{ccccc} B & \xrightarrow{\iota_B} & \mathbf{P}^N \times \{[B]\} & \longrightarrow & \{[B]\} \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{A} & \xrightarrow{\iota} & \mathbf{P}_{\mathcal{M}}^N & \xrightarrow{\text{pr}_2} & \mathcal{M} \end{array} .$$

Then (3.4) implies that $(R\iota^!L)|_B \simeq R\iota_B^!Rf_*K$. Coupled with the purity isomorphism (3.3), we get

$$\iota_B^{-1}Rf_*K[-2](-1) \simeq R\iota_B^!Rf_*K.$$

Now we turn to the following fiber diagram:

$$\begin{array}{ccc} B \times_{\mathbf{P}^N} X & \xrightarrow{\iota'_B} & X \\ \downarrow f' & & \downarrow f \\ B & \xrightarrow{\iota_B} & \mathbf{P}^N \end{array} .$$

By proper base change, we see $\iota_B^{-1}Rf_*K \simeq Rf'_*(\iota'_B)^{-1}K$. Applying Verdier duality to the above isomorphism gives $R\iota_B^!Rf_!\mathbf{D}K \simeq Rf'_!R(\iota'_B)^!\mathbf{D}K$. Here $\mathbf{D}K$ is the Verdier dual of K . Replacing K by $\mathbf{D}K$ in this latter isomorphism, using $Rf_! = Rf_*$ (as f is proper) and $Rf'_! = Rf'_*$, we conclude that

$$Rf'_*(\iota'_B)^{-1}K[-2](-1) \simeq Rf'_*R(\iota'_B)^!K. \quad (3.5)$$

Since f is a finite morphism, so is f' . In this situation, we have $Rf'_* = f'_*$. For any closed point b of B , the stalks of both sides of (3.5) at b are the direct sums of the stalks of $(\iota'_B)^{-1}K[-2](-1)$ and $R(\iota'_B)^!K$ at the points in $f^{-1}(b)$. Since the purity map induces an isomorphism on the direct sum of these stalks, the purity map itself is necessarily an isomorphism. Therefore, (3.5) implies that $K|_{B \times_{\mathbf{P}^N} X}[-2](-1) \simeq R(\iota'_B)^!K$. This finishes the proof when f is a finite morphism.

Now assume f is a quasi-finite morphism. Then we can factor f into a composition $X \xrightarrow{j} \overline{X} \xrightarrow{g} \mathbf{P}^N$ in which j is an open immersion, g is a finite morphism. Thus we have the following fiber diagram

$$\begin{array}{ccc} X \times_{\mathbf{P}^N} B & \xrightarrow{\iota'_B} & X \\ \downarrow j' & & \downarrow j \\ \overline{X} \times_{\mathbf{P}^N} B & \xrightarrow{\iota''_B} & \overline{X} \\ \downarrow g' & & \downarrow g \\ B & \xrightarrow{\iota_B} & \mathbf{P}^N \end{array} .$$

In the diagram, $f' = g' \circ j'$. By the previous paragraph, for a sufficiently general B , we have

$$(j_*K)|_{\overline{X} \times_{\mathbf{P}^N} B}[-2](-1) \simeq R(\iota''_B)^!(j_*K).$$

Applying the functor $(j')^{-1} = R(j')^!$ to the above isomorphism gives the desired isomorphism

$$K|_{X \times_{\mathbf{P}^N} B}[-2](-1) \simeq R(\iota'_B)^!K.$$

This completes the proof. ■

Next, let us recall Deligne's weak Lefschetz theorem for perverse sheaves [Kat93, Corollary A.5].

Theorem 3.6 (Deligne). *Let $f: X \rightarrow \mathbf{P}^N$ be a quasi-finite morphism to a projective space. Let K be a perverse sheaf on X . Then for a sufficiently general hyperplane B , the restriction morphism*

$$H^i(X; K) \rightarrow H^i(f^{-1}B; K|_{f^{-1}B})$$

is injective if $i = -1$, and bijective if $i < -1$.

Remark 3.7. Going through the proof of Theorem 3.6, one finds that in the statement one may replace the perversity of K by K satisfying the *cosupport condition*. Recall that a constructible complex K is said to be *semiperverse*, or to satisfy the *support condition*, if $\dim \text{Supp } \mathcal{H}^i(K) \leq -i$. In terms of the perverse t -structure, this means that K has vanishing positive perverse cohomology sheaves. If $\mathbf{D}K$ satisfies the support condition then we say K satisfies the *cosupport condition*.

Lemma 3.8. *Let $f: X \rightarrow \mathbf{P}^N$ be a quasi-finite morphism. Let K be a perverse sheaf on X (or more generally, K satisfies the support condition). Then for a sufficiently general hyperplane, and any i , there exists a Gysin map*

$$H_c^{i-2}(X \times_{\mathbf{P}^N} B; K|_{X \times_{\mathbf{P}^N} B}(-1)) \rightarrow H_c^i(X; K),$$

which is surjective if $i = 1$, and is bijective if $i \geq 2$.

Proof. We choose a sufficiently generic hyperplane B such that the conclusion of Theorem 3.6 holds for $\mathbf{D}K$, and such that the conclusion of Lemma 3.2 holds for K . Thus

$$H^i(X; \mathbf{D}K) \rightarrow H^i(X \times_{\mathbf{P}^N} B; \iota_B^*(\mathbf{D}K))$$

is injective for $i = -1$, and is bijective for $i < -1$. Taking Verdier duality yields that the natural map

$$H_c^i(X \times_{\mathbf{P}^N} B; R\iota_B^!K) \rightarrow H_c^i(X; K)$$

is surjective for $i = 1$, and is bijective for $i > 1$. By our choice, $R\iota_B^! K \simeq \iota_B^* K[-2](-1)$. Therefore we obtain a map

$$H_c^{i-2}(X \times_{\mathbf{P}^N} B; K|_{X \times_{\mathbf{P}^N} B}(-1)) \rightarrow H_c^i(X; K),$$

which is surjective for $i = 1$, and bijective for $i > 1$. ■

Let X be an algebraic variety of dimension $\leq n$. Then for any constructible Λ -sheaf $\mathcal{F}$ on X , it is trivial to show that $\mathcal{F}[n]$ satisfies the support condition. Applying Lemma 3.8 gives the following result.

Lemma 3.9. *Let X be a k -variety of dimension $\leq n$. Let $f: X \rightarrow \mathbf{P}^N$ be a quasi-finite immersion. Let $\mathcal{F}$ be a constructible sheaf on X . Then for a sufficiently general hyperplane B , there exists a Gysin map*

$$H_c^{i-2}(X \times_{\mathbf{P}^N} B; \mathcal{F}(-1)) \rightarrow H_c^i(X; \mathcal{F})$$

which is surjective if $i = n + 1$, and is bijective if $i \geq n + 2$. ■

4 The method of del Angel and Esnault–Katz

Situation 4.1. Let $f_1, \dots, f_r \in k[x_1, \dots, x_N]$ be a collection of polynomials. Let X be the common zero locus of $f_1, \dots, f_r$ in $\mathbf{A}^N$. Assume that $d_1 \geq \dots \geq d_r$ is a sequence of numbers such that $d_i \geq \deg f_i$. Let $n = \dim X$. In general, $n \geq N - r$; the equality holds if and only if X is a (set-theoretic) complete intersection by the r defining equations.

The following lemma proves the main theorems when X is a set-theoretic complete intersection, and it also proves Theorem 1.12.

Lemma 4.2. *In Situation 4.1, the colevel of $H_c^{n+j}(X)$ is $\geq \mu_j(N; d_1, \dots, d_r)$ for $j \geq 0$.*

Proof. Choose n sufficiently general hyperplanes $B_1, \dots, B_n$. Since $B_1, \dots, B_n$ are general, for each j , $X \cap B_1 \cap \dots \cap B_j$ is a Zariski closed subset cut out in $\mathbf{A}^{N-j}$ by equations of degrees at most $d_1, \dots, d_r$, and has dimension equal to $n - j$.

Applying the Gysin lemma, Lemma 3.9, to X_i gives the following chain of maps:

$$\begin{array}{ccccccc} H_c^n(X) & & H_c^{n+1}(X) & & H_c^{n+2}(X) & & \dots \\ & & \uparrow & & \uparrow & & \\ & & H_c^{n-1}(X \cap B_1; \Lambda(-1)) & & H_c^n(X \cap B_1; \Lambda(-1)) & & \dots \\ & & & & \uparrow & & \\ & & & & H_c^{n-2}(X \cap B_1 \cap B_2; \Lambda(-2)) & & \dots \\ & & & & & & \ddots \end{array}$$

It follows that the colevel of $H_c^{n+j}(X)$ is at least the colevel of $H_c^{n-j}(X \cap B_1 \cap \dots \cap B_j)$ plus j . By the theorem of Esnault and Katz, Theorem 1.10(1a), the colevel of $H_c^{n-j}(X \cap B_1 \cap \dots \cap B_j)$ is at least $\mu_0(N - j; d_1, \dots, d_r)$. It follows that the colevel of $H_c^{n+j}(X)$ is at least $\mu_0(N - j; d_1, \dots, d_r) + j = \mu_j(N; d_1, \dots, d_r)$. ■

While Lemma 4.2 answers positively Question 1.11, it does not provide the most optimal bounds unless $n = N - r$, and it does not provide information for colevels of H_c^j when $N - r \leq j \leq n$. In the following, we shall focus on proving the improved bounds stated in Theorems 2.2 and 2.3.

We shall first treat the case when X is “not too far from” a complete intersection; that is, X is obtained from a complete intersection by adding an extra equation which does not drop the dimension. See Lemma 4.7. Note however we formulate the result in a slightly more general way. This is to facilitate an induction argument used in next section. This sort of formulation had been used by del Angel [dAR94] and Esnault–Katz. Let us state the theorem of Esnault–Katz [EK05, Theorem 2.1, §5.3], which is slightly stronger than Theorem 1.10(1a), and will be used later.

Theorem 4.3 (del Angel, Esnault–Katz). *Suppose we are given a list of polynomials $f_1, \dots, f_r \in k[x_1, \dots, x_N]$ with $\deg f_i \leq d_i$, $d_i \geq 1$. Take any integer $R \geq 1$ and any list of polynomials $g_1, \dots, g_R \in k[x_1, \dots, x_N]$, such that each g_j is a product $g_j = \prod_i f_i^{a_{i,j}}$ with $a_{i,j} \geq 0$. Denote by U the closed subscheme of $\mathbf{A}^N$ defined by $g_1 = \dots = g_R = 0$. Then for any $m \in \mathbf{Z}$, the colevel of $H_c^m(U)$ is at least $\mu_0(N; d_1, \dots, d_r)$.*

Situation 4.4. In Situation 4.1, let V be the variety defined by $f_1 = \dots = f_{N-n} = 0$. We make the following additional hypothesis:

$$\dim V = \dim X = n.$$

In other words, we assume in addition that V is a set-theoretic complete intersection cut out by $f_1, \dots, f_{N-n}$.

For every nonempty subset J of $\{N - n + 1, \dots, r\}$, let X_J be the closed subscheme of V defined by $\prod_{j \in J} f_j = 0$. Since $V \supset X_J \supset X$, we know $\dim X_J = n$. Note that when J is the empty set, we have $X_J = V$ by convention.

Lemma 4.5. *In Situation 4.4, for every $J \subset \{N - n + 1, \dots, r\}$, the colevel of $H_c^n(V - X_J)$ is $\geq \mu_0(N; d_1, \dots, d_r)$.*

Proof. Since V is an affine complete intersection of pure dimension n , and since X_J is a hypersurface in V , the open subscheme $V - X_J$ is an affine local complete intersection of pure dimension n . Consider the following long exact sequence

$$\begin{array}{ccccccc} 0 & \longrightarrow & 0 & \longrightarrow & H_c^{n-1}(X_J) & \longrightarrow & \\ & & & & \downarrow & & \\ & & H_c^n(V - X_J) & \longrightarrow & H_c^n(V) & \longrightarrow & H_c^n(X_J) \\ & & & & \downarrow & & \\ & & H_c^{n+1}(V - X_J) & \longrightarrow & H_c^{n+1}(V) & \longrightarrow & H_c^{n+1}(X_J) \longrightarrow \dots \end{array} \quad (4.6)$$

By Lemma 4.2, the colevel of $H_c^n(V)$ is at least $\mu_0(N; d_1, \dots, d_{N-n})$. By Theorem 4.3 ($R = N - n + 1$, $U = X_J = \{f_1 = \dots = f_{N-n} = \prod_{j \in J} f_j = 0\}$), the colevel of $H_c^{n-1}(X_J)$ is at least $\mu_0(N; d_1, \dots, d_r)$. From the exact sequence (4.6), we see the colevel of $H_c^n(V - X)$ is at least $\mu_0(N; d_1, \dots, d_r)$. ■

Lemma 4.7. *Assume $r \geq N - n + 1$. In Situation 4.4, for every subset J of $\{N - n + 1, \dots, r\}$, the colevel of $H_c^{n+j}(X_J)$ is at least $\nu_j^{(r-1)}(N; d_1, \dots, d_r)$.*

Proof. From the exact sequence (4.6) we see that the colevel of $H_c^{n+j}(X_J)$ is controlled by $H_c^{n+j}(V)$ and $H_c^{n+j+1}(V - X_J)$. The colevel of the former is at least $\mu_j(N; d_1, \dots, d_{N-n})$, as shown in Lemma 4.2. Clearly

$$\frac{N - j - \sum_{i=1}^{N-n} d_i}{d_1} \geq \frac{N - j - \sum_{i=1}^r d_i^*(e)}{d_1},$$

for every $e \geq N - n$. Taking $e = N - n \leq r - 1$, we conclude that

$$\mu_j(N; d_1, \dots, d_{N-n}) \geq \nu_j^{(N-n)}(N; d_1, \dots, d_r) \geq \nu_j^{(r-1)}(N; d_1, \dots, d_r).$$

It remains to prove the colevel of $H_c^{n+j+1}(V - X_J)$ is $\geq \nu_j^{(r-1)}(N; d_1, \dots, d_r)$.

For each j , choose a sufficiently general linear subspace B in $\mathbf{A}^N$ of codimension $j + 1$. The Gysin lemma, Lemma 3.9, applied to the locally closed immersion

$$V - X_J \hookrightarrow \mathbf{A}^N \subset \mathbf{P}^N$$

($j + 1$ times), gives a surjection

$$H_c^{n-(j+1)}((V \cap B) - (X_J \cap B); \Lambda(-j-1)) \rightarrow H_c^{n+j+1}(V - X_J).$$

Now, $V \cap B$ is a complete intersection of dimension $n - (j + 1)$ in $\mathbf{A}^{N-(j+1)}$. By Lemma 4.5, we deduce that the colevel of $H_c^{n+j+1}(V - X_J)$ is at least

$$j + 1 + \mu_0(N - j - 1; d_1, \dots, d_r) = \mu_{j+1}(N; d_1, \dots, d_r).$$

Since $r \geq N - n + 1 > 1$, Lemma 2.7(3) can be applied, and we see the above number is $\geq \nu_j^{(r-1)}(N; d_1, \dots, d_r)$. ■

5 Proofs of main theorems

In this section we prove Theorems 2.2 and 2.3. It suffices to prove the colevel bounds for the affine variety X . The colevel bounds for the complement $\mathbf{A}^N - X$ then follows by excision.

We will need the following simple lemma.

Lemma 5.1. *Let W be an algebraic variety. Let $W_1, \dots, W_s$ be a collection of Zariski closed subsets of W , such that $W = \bigcup_{v=1}^s W_v$. Then the colevel of $H_c^i(W_1 \cap \dots \cap W_s)$ is at least the minimum of the colevels of the following cohomology groups:*

$$\bigoplus_{|I|=s-a} H_c^{i+a-1} \left(\bigcap_{v \in I} W_v \right), \quad a = 1, 2, \dots, s.$$

We remark that when $a = s$, the intersection $\bigcap_{v \in I} W_v$, $|I| = s - a$, is an empty intersection, which corresponds to the entire set W . So the boundary case $a = s$ of the statement corresponds to $H_c^{i+s-1}(W)$.

Proof. There is a first-quadrant Mayer–Vietoris spectral sequence

$$E_1^{a,b} = \bigoplus_{|I|=a+1} H_c^b \left(\bigcap_{v \in I} W_v \right) \Rightarrow H_c^{a+b}(W)$$

associated to the finite proper hypercovering $W = \bigcup W_\nu$. In the ℓ -adic context, the differentials preserve the geometric Frobenius operation; in the Betti context, the differentials are morphisms of mixed Hodge structures.

To explain the idea of the proof, we have drawn below a schematic diagram, Figure 1. In the figure, the gray squares represent the positions in the spectral sequence that may hit the terms sitting in the red square, which are $E_\bullet^{s-1,j}$. A certain subquotients of the first-page items sitting in the gray squares will span a linear subspace of $E_1^{s-1,j}$. Dividing by this subspace gives a quotient space of $E_1^{s-1,j}$, which equals $E_\infty^{s-1,j}$ and is also a quotient of the cohomology of W . Therefore, to estimate the lower bound of the colevel of the first-page item at the red square, that is $E_1^{s-1,*} = H_c^*(\bigcap_{\nu=1}^s W_\nu)$, we only need some lower bounds of the colevels of the first-page items sitting in the gray squares, and a lower bound of the colevel of $H_c^{s-1+*}(W)$.

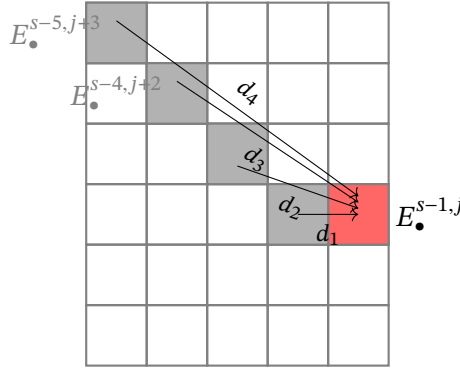


Figure 1: The spectral sequence

Now we give the formal proof. The cohomology group $H_c^*(W_1 \cap \dots \cap W_s)$ in question equals $E_1^{s-1,*}$ (indicated by the red box). As it is in the right-most column of the spectral sequence, all the differentials out of it are zero. It follows that $E_\infty^{s-1,*}$ is a quotient of $E_1^{s-1,*}$. We have

$$E_2^{s-1,*} = E_1^{s-1,*} / d_1(E_1^{s-2,*}), \dots, E_{a+1}^{s-1,*} = E_a^{s-1,*} / d_a(E_a^{s-a-1,*+a-1}), \dots,$$

and $E_\infty^{s-1,*} = E_s^{s-1,*}$. Since each item $E_a^{s-a-1,*+a-1}$ is a subquotient of

$$E_1^{s-a-1,*+a-1} = \bigoplus_{|I|=s-a} H_c^{*+a-1} \left(\bigcap_{\nu \in I} W_\nu \right)$$

(indicated by the gray boxes), it follows that the colevel of the kernel of $E_1^{s-1,*} \rightarrow E_\infty^{s-1,*}$ is at least the minimum of

$$\bigoplus_{|I|=s-a} H_c^{*+a-1} \left(\bigcap_{\nu \in I} W_\nu \right), \quad a = 1, 2, \dots, s-1.$$

To complete the proof, it remains to bound the colevel of $E_\infty^{s-1,*}$. Since $E_\infty^{s-1,*}$ is a subspace of the abutment $H_c^{*+s-1}(W)$ (the $(s-1)^{\text{st}}$ column is the right-most column), its colevel is at least the colevel of $H_c^{*+s-1}(W)$. This completes the proof. $\blacksquare$

We also need the following crucial algebraic lemma. It allows us to rebase the problem from $\mathbf{A}^N$ to a set-theoretic complete intersection of dimension n . Since the proof is a bit lengthy, we shall not reproduce it here. The reader is referred to [WZ25, Lemma 5.1].

Lemma 5.2. *Let k be a field. Let $f_1, \dots, f_r \in k[x_1, \dots, x_N]$ be a collection of polynomials. Assume that $\deg f_1 \geq \dots \geq \deg f_r$. Let X be the affine variety defined by $f_1 = \dots = f_r = 0$ in $\mathbf{A}^N$. Then, upon replacing k by a finite extension, there exists a new sequence of polynomials $g_1, \dots, g_r \in k[x_1, \dots, x_N]$ such that*

1. $X = \{g_1 = \dots = g_r = 0\}$,
2. $\deg g_i \leq \deg f_i$,
3. *the dimension of the variety V defined by $\{g_1 = \dots = g_{N-\dim X} = 0\}$ is $\dim X$.*

Proof of Theorem 2.2. Recall the definition of the numbers $\nu_j^{(e)}(N; d_1, \dots, d_r)$ in Definition 2.1. By Lemma 5.2, we can replace f_i by g_i without changing X . Since $\nu_j^{(e)}$ is decreasing in d_j (Lemma 2.7(2)), the inequality “ $\text{colevel} \geq \nu_j^{(e)}(\deg g_1, \dots, \deg g_r)$ ” is stronger than the desired inequality. For this reason, we may replace f_i by g_i , and it suffices prove the theorem under the following hypothesis:

$$\text{The variety } V \text{ defined by } f_1 = \dots = f_{N-n} = 0 \text{ has dimension } n. \quad (5.3)$$

Under hypothesis (5.3), X is a subvariety of V defined by $r - (N - n)$ “extra equations”; these equations do not drop the dimension: $\dim X = \dim V$.

We shall prove the theorem by induction on $r - (N - n)$, the number of extra equations needed to cut out X in V . When this number is 0, the theorem is Lemma 4.2. Assume now there exist $r - (N - n) > 0$ “extra equations”. The theorem is already true if $r - (N - n) = 1$ by Lemma 4.7. But, we will need the more general form of Lemma 4.7 in the induction below.

For $i \in \{N - n + 1, \dots, r\}$, let $X_i = V \cap \{f_i = 0\}$. Then $X = X_{N-n+1} \cap \dots \cap X_r$. In view of Lemma 5.1 (with $s = r - (N - n)$), it suffices to prove

for each $I \subset \{N - n + 1, \dots, r\}$ such that $|I| = r - (N - n) - a$, $1 \leq a \leq r - (N - n)$, the colevel of $H_c^{n+j+a-1}(\cap_{i \in I} X_i)$ is at least $\nu_j^{(N-n)}(N; d_1, \dots, d_r)$.

Situation I. $1 \leq a \leq r - (N - n) - 1$.

This case is handled by the inductive hypothesis. The intersection $\cap_{i \in I} X_i$ is a subvariety of V of dimension n , which is cut out by $r - (N - n) - a < r - (N - n)$ equations. The inductive hypothesis implies that $H_c^{n+j+a-1}(\cap_{i \in I} X_i)$ has $\text{colevel} \geq \nu_{j+a-1}^{(N-n)}(d_1, \dots, d_{r-a})$. By Lemma 2.7(4), this number is $\geq \nu_j^{(N-n)}(N; d_1, \dots, d_r)$.

Situation II. $a = r - (N - n)$. This is the case when the intersection is empty. Let $W = X_{N-n+1} \cup \dots \cup X_r$. We need to show the colevel of $H_c^{n+j+r-(N-n)-1}(W)$ is at least

$$\nu_j^{(N-n)}(N; d_1, \dots, d_r).$$

This is handled by Lemma 4.7: The variety W is defined by the equation $f_{N-n+1} \dots f_r = 0$, and satisfies the hypothesis of the lemma. Hence the colevel of $H_c^{n+j+r-(N-n)-1}(X)$ is at least

$$\nu_{j+r-(N-n)-1}^{(r-1)}(N; d_1, \dots, d_r).$$

Since $r > 0$, we have $N - n > 0$. It follows that $r - 1 > r - (N - n) - 1$. We can then apply Lemma 2.7(3), and conclude that

$$\begin{aligned} \nu_{j+r-(N-n)-1}^{(r-1)}(N; d_1, \dots, d_r) &\geq \nu_j^{(r-1-(r-(N-n)-1))}(N; d_1, \dots, d_r) \\ &= \nu_j^{(N-n)}(N; d_1, \dots, d_r). \end{aligned}$$

This completes the proof. $\blacksquare$

Proof of Theorem 2.3. As before, without loss of generality, we may assume the additional hypothesis (5.3) holds. We shall prove by induction on $r - (N - n)$, the number of “extra equations” that are needed to cut out X from the complete intersection V .

The base case is when $N - n = r$, i.e., when $X = V$ is a set-theoretic complete intersection. In this case, Theorem 2.3 falls back to Lemma 4.2.

Now assume $r > N - n$. If $m = 0$, then we are reduced to the theorem of Esnault and Katz (Theorem 1.10(1a)). When $m > 0$, by Lemma 5.1 ($s = r - (N - n)$), it suffices to show, that

for each subset I of $\{N - n + 1, \dots, r\}$ such that $|I| = r - (N - n) - a$ for some $1 \leq a \leq r - (N - n)$, and each integer m such that $0 \leq m \leq r - (N - n)$, the colevel of $H_c^{N-r+m+a-1}(\cap_{i \in I} X_i)$ is at least $\nu_0^{(r-m)}(N; d_1, \dots, d_r)$.

Again, this is an elementary consequence of Lemma 2.7.

Situation I. $1 \leq a \leq r - (N - n) - 1$.

There are two subcases:

1. $N - r + m + a - 1 < n$, i.e., when the cohomological degree in question is less than n ; and
2. $N - r + m + a - 1 \geq n$, i.e., when the cohomological degree is larger than or equal to n .

Case (1) is handled by induction. In this case, each $\cap_{i \in I} X_i$ is cut out in $\mathbf{A}^N$ by $(r - a)$ total defining equations, and there are $(r - a) - (N - n)$ “extra equations”. Write $N - r + m + a - 1 = N - (r - a) + m - 1$. The inductive hypothesis shows that the colevel of $H_c^{N-r+m+a-1}(\cap_{i \in I} X_i)$ is at least $\geq \nu_0^{((r-a)-(m-1))}(N; d_1, \dots, d_{r-a})$. By Lemma 2.7(2), this number is $\geq \nu_0^{(r-m)}(N; d_1, \dots, d_{r-a})$. This latter number is trivially $\geq \nu_0^{(r-m)}(N; d_1, \dots, d_r)$.

Case (2) is handled by Theorem 2.2. Since $N - r + m + a - 1 \geq n$, we write it as $n + (N - n - r + m + a - 1)$. By Theorem 2.2, a lower bound of the colevel of $H_c^{N-r+m+a-1}(\cap_{i \in I} X_i)$ is $\nu_{N-n-r+m+a-1}^{(N-n)}(N; d_1, \dots, d_{r-a})$. By Lemma 2.7(3), this number is $\geq \nu_0^{(r-m-a+1)}(N; d_1, \dots, d_r)$. Since $0 \geq 1 - a$, and since $r \geq r - (N - n) \geq m$ by assumption, $r - m - a + 1 \geq 1$. Thus, Lemma 2.7(2) implies that the above number is $\geq \nu_0^{(r-m)}(N; d_1, \dots, d_r)$.

Situation II. $a = r - (N - n)$.

This is when the intersection is empty, so we need to show the colevel of $H_c^{n+m-1}(\cup X_i)$ is at least $\nu_0^{(r-m)}(N; d_1, \dots, d_r)$. When $m = 0$, this is the Ax-Katz bound, so we are reduced to Theorem 1.10(1a). If $m \geq 1$, by Lemma 4.7, the colevel of $H_c^{n+m-1}(\cup X_i)$ is at least $\nu_{m-1}^{(r-1)}(N; d_1, \dots, d_r)$. By Lemma 2.7(3), this number is $\geq \nu_0^{((r-1)-(m-1))}(N; d_1, \dots, d_r) = \nu_0^{(r-m)}(N; d_1, \dots, d_r)$, as desired. This completes the proof. $\blacksquare$

Proof of Theorem 2.5. It suffices to prove the claim about cohomology of Y ; the claim for $\mathbf{P}^N - Y$ follows from the usual long exact sequence for relative cohomology. We shall prove the theorem by induction on $n = \dim Y$. The base case ($n = 0$) is trivial. If $n > 0$ and $n = N$, then the theorem is also trivial. So we may assume $0 < n < N$.

Let $\widehat{Y} \subset \mathbf{A}^{N+1}$ be the affine cone of Y . Let $L \rightarrow Y$ be the total space of the “tautological line bundle” of $\mathbf{P}^N$ restricted to Y . Thus, Y embeds into L as the zero section, whose complement can be identified with $\widehat{Y} - O$. We have a long exact sequence

$$\cdots \rightarrow H_c^i(L) \xrightarrow{u} H^i(Y) \rightarrow H_c^{i+1}(\widehat{Y} - O) \rightarrow H_c^{i+1}(L) \rightarrow \cdots \quad (5.4)$$

Since the fibers of $L \rightarrow Y$ are $\mathbf{A}^1$, its Leray spectral sequence degenerates into an isomorphism $H_c^i(L) \simeq H^{i-2}(Y; \Lambda(-1))$. Under this isomorphism, the mapping u is identified with the cup product with c_1 mapping

$$\smile c_1(\mathcal{O}_Y(-1)): H^{i-2}(Y; \Lambda(-1)) \rightarrow H^i(Y; \Lambda).$$

Moreover, for any $i \geq 1$, the natural morphism

$$H_c^{i+1}(\widehat{Y} - O) \xrightarrow{\sim} H_c^{i+1}(\widehat{Y})$$

is an isomorphism. Therefore, for $i \geq 1$, the exact sequence (5.4) can be rewritten as

$$\cdots \rightarrow H^{i-2}(Y; \Lambda(-1))_{\text{prim}} \xrightarrow{\smile c_1(\mathcal{O}_Y(1))} H^i(Y)_{\text{prim}} \rightarrow H_c^{i+1}(\widehat{Y}) \rightarrow \cdots \quad (5.5)$$

Here, we have made two small modifications. First, we have decided to take the cup product with $c_1(\mathcal{O}_Y(1)) = -c_1(\mathcal{O}_Y(-1))$. Obviously this does not affect the exactness. Second, we have taken the primitive part. This modification preserves exactness because the non-primitive classes are precisely the powers of $c_1(\mathcal{O}_Y(1))$.

By Theorems 2.2 and 2.3,

- if $i = N - r + m$ for some $0 \leq m \leq r - (N - n)$, then the colevel of $H_c^{i+1}(\widehat{Y})$ is at least $\nu_0^{(r-m)}(N + 1; d_1, \dots, d_r)$;
- if $i = n + j$, for some $j \geq 1$, then the colevel of $H_c^{i+1}(\widehat{Y})$ is at least $\nu_j^{(N-n)}(N + 1; d_1, \dots, d_r)$.

Thus, by the exactness of (5.5), it suffices to prove that the colevel of the image of the mapping

$$H^{i-2}(Y; \Lambda(-1))_{\text{prim}} \xrightarrow{\smile c_1(\mathcal{O}_Y(1))} H^i(Y)_{\text{prim}} \quad (5.6)$$

is at least

- $\nu_0^{(r-m)}(N + 1; d_1, \dots, d_r)$, if $i = N - r + m$ for some $1 \leq m \leq r - (N - n)$ (the $m = 0$ case already being taken care of by Theorem 1.10(2a)), and
- $\nu_j^{(N-n)}(N + 1; d_1, \dots, d_r)$, if $i = n + j$ for some $j \geq 1$.

Let A be any hyperplane in $\mathbf{P}^N$. We claim that the cup product mapping be factored as

$$\begin{array}{ccccc} H^{i-2}(Y; \Lambda(-1)) & \longrightarrow & H^{i-2}(Y \cap A; \Lambda(-1)) & \longrightarrow & H^i(Y) \\ & & \searrow \smile c_1(\mathcal{O}_Y(1)) & \nearrow & \\ & & & & \end{array} \quad (5.7)$$

where the left arrow is the restriction map. For the moment, let us grant the existence of the factorization (5.7), and see how it implies the theorem. Since A can be arbitrarily chosen, we shall choose it so that $\dim Y \cap A = \dim Y - 1 = n - 1$. Viewing A as an $N - 1$ dimensional projective space, we can apply the inductive hypotheses to the $(n - 1)$ -dimensional scheme $Y \cap A$. Therefore, the colevel of $H^{i-2}(Y \cap A; \Lambda(-1))_{\text{prim}}$ is at least

- (α) $1 + \nu_0^{(r-(m-1))}(N+1; d_1, \dots, d_r)$ if $i = N - r + m$ (that is, $i - 2 = (N - 1) - r + (m - 1)$), for $1 \leq m \leq r - (N - n)$;
- (β) $1 + \nu_{j-1}^{((N-1)-(n-1))}(N; d_1, \dots, d_r)$, if $i = n + j$ (that is, $i - 2 = (n - 1) + (j - 1)$), $j \geq 1$.

The induction proof is concluded by observing the following elementary facts:

- (0) (5.7) gives a factorization

$$H^{i-2}(Y; \Lambda(-1))_{\text{prim}} \rightarrow H^{i-2}(Y \cap A; \Lambda(-1))_{\text{prim}} \rightarrow H^i(Y)_{\text{prim}};$$

of (5.6)

- (1_e) $1 + \nu_0^{(e+1)}(N; d_1, \dots, d_r) \geq \nu_0^{(e)}(N+1; d_1, \dots, d_r)$, for any $e \in \mathbf{Z}$, and
- (2_e) $1 + \nu_{j-1}^{(e)}(N; d_1, \dots, d_r) = \nu_j^{(e)}(N+1; d_1, \dots, d_r)$ for any $j \geq 1$.

Indeed, given Item (0), (α) and (1_{r-m}) imply (a), and (β) and (2_{N-n}) imply (b). Item (0) is straightforward, and Item (2) can be seen simply by expanding the definition. To prove (1), we note that by (2), we have $1 + \nu_0^{(e+1)}(N; d_1, \dots, d_r) = \nu_1^{(e)}(N; d_1, \dots, d_r)$. Then (1) follows from Lemma 2.7(3). ■

It remains to explain the existence of the factorization (5.7).

Existence of (5.7). The Chern class $c_1(\mathcal{O}_{\mathbf{P}^N}(1)) \in H^2(\mathbf{P}^N; \Lambda(1))$ can be regarded as a mapping, in the derived category of constructible complexes of Λ -sheaves

$$\Lambda_{\mathbf{P}^N}(-1)[-2] \xrightarrow{\sim c_1(\mathcal{O}_{\mathbf{P}^N}(1))} \Lambda_{\mathbf{P}^N}. \quad (5.8)$$

Let A be a hyperplane in $\mathbf{P}^N$, and let $\iota: A \rightarrow \mathbf{P}^N$ be the inclusion morphism. The relative purity theorem implies that $\iota^! \Lambda_{\mathbf{P}^N} = \Lambda_A(-1)[-2]$. From this, we obtain a “Gysin morphism” $\iota_* \Lambda_A(-1)[-2] \xrightarrow{\text{Gys}} \Lambda_{\mathbf{P}^N}$ arising from the counit morphism $\iota_* \iota^! \Lambda_{\mathbf{P}^N} \rightarrow \Lambda_{\mathbf{P}^N}$. The mapping (5.8) can be identified with

$$\begin{array}{ccc} \Lambda_{\mathbf{P}^N}(-1)[-2] & \longrightarrow & \iota_* \Lambda_A(-1)[-2] \xrightarrow{\text{Gys}} \Lambda_{\mathbf{P}^N} \\ & \searrow \sim c_1(\mathcal{O}_{\mathbf{P}^N}(1)) \nearrow & \end{array} \quad (5.9)$$

This identification holds because the cycle class of A coincides with $c_1(\mathcal{O}_{\mathbf{P}^N}(1))$.

Since taking the Chern class is compatible with the restriction of invertible sheaves, applying the $*$ -inverse image to (5.8) with respect to $Y \hookrightarrow \mathbf{P}^N$ gives rise to the Chern class of $\mathcal{O}_Y(1)$. By restricting (5.9) to Y and applying the proper base change theorem, we obtain the following factorization:

$$\begin{array}{ccc} \Lambda_Y(-1)[-2] & \longrightarrow & \iota'_* \Lambda_{Y \cap A}(-1)[-2] \xrightarrow{\text{Gys}} \Lambda_Y, \\ & \searrow \sim c_1(\mathcal{O}_Y(1)) \nearrow & \end{array} \quad (5.10)$$

where $\iota': Y \cap A \rightarrow Y$ denotes the inclusion morphism.

Applying $R\Gamma(Y; -)$ to the sheaf-theoretic factorization (5.10) gives the cohomological factorization (5.7), as claimed. $\blacksquare$

Remark 5.11. A version of this proof appeared in [WZ25]. However, the argument there contained a gap: it incorrectly claimed that the image of u in (5.4) was always “non-primitive”. The proof presented above corrects this oversight: although the image of u may indeed be primitive, it arises as the Gysin image of a hyperplane section, whose colevel can be controlled through induction. We thank an anonymous referee for bringing this issue to our attention.

6 Two examples

We end with two “determinantal” examples illustrating how the new bounds provided by Theorems 2.2 and 2.3 are sharper than the previously known (or anticipated) bounds.

Example 6.1. Let e be a positive integer ≥ 4 . Let $N = e^2$. View $\mathbf{A}^N$ as the space of $e \times e$ -matrices. Consider the following variety

$$X = \{A \in \mathbf{A}^N : \text{all principal } (e-1)\text{-minors are zero}\}.$$

Then X is defined by the vanishing of $r = e$ equations of degree $e-1$. The Ax–Katz bound implies that the colevels of $H_c^*(X)$ are ≥ 2 .

According to Wheeler [Whe15, Theorem 4], X has two irreducible components: $X = X_1 \cup X_2$, where $X_1 = \{A \in \mathbf{A}^N : \text{rank } A \leq e-2\}$, X_2 is the Zariski closure of $\{A \in X : \det A \neq 0\}$. Moreover, $\dim X_2 = N - e$. On the other hand, it is well-known that $\dim X_1 = e^2 - 4$ ([ACGH85, p. 67, Proposition]). Thus the dimension of X is $n = e^2 - 4$. The affine variety X in $\mathbf{A}^N$ is a complete intersection if $e = 4$, not a complete intersection if $e \geq 5$.

The improved bounds, Theorem 2.2 and 2.3, give the following:

- (before middle dimension) for $0 \leq m \leq e-4$, the colevel of $H_c^{e^2-e+m}(X)$ is at least $\lceil m+1 - \frac{m-1}{e-1} \rceil$ which is $m+1$ if $1 \leq m \leq e-4$, and 2 if $m = 0$ (Theorem 1.10 only shows that these colevels are ≥ 2);
- (beyond middle dimension) for $0 \leq j \leq e^2 - 4$, the colevel of $H_c^{e^2-4+j}(X)$ is at least $j + \max\left\{0, e-4 + \left\lceil \frac{4-j}{e-1} \right\rceil\right\}$. When $e \geq 5$, this latter value is $j+e-3$ if $0 \leq j \leq 3$, $j+e-4$ if $4 \leq j \leq e+2$, $j+e-5$ if $e+3 \leq j \leq 2e+1$, etc. By contrast, the colevel lower bound anticipated by Esnault and the first author (Theorem 1.12) is $j + \max\left\{0, \left\lceil \frac{e-j}{e-1} \right\rceil\right\}$, which is $j+2$ if $j = 0$, $j+1$ if $1 \leq j \leq e-1$ and j if $j \geq e$.

Example 6.2. Notation as Example 6.1, consider

$$X' = \{A \in \mathbf{A}^N : \text{rank } A \leq e-1, \text{ all principal } (e-1)\text{-minors are zero}\}.$$

Then X' is cut out by $r = e+1$ equations: the vanishing of the e principal minors (degree $e-1$ polynomials), and $\det A = 0$ (a degree e polynomial). Thus $d_1 = e, d_2 = \dots = d_r = e-1$.

From Wheeler’s theorem cited above it is easy to see that we can write $X' = X_1 \cup X'_2$, where $X'_2 = X_2 \cap \{\det = 0\}$. Thus X'_2 has codimension $e+1 \geq 5$ in $\mathbf{A}^N$. It follows that X' has codimension 4 in $\mathbf{A}^N$.

Since $\frac{N-\sum d_i}{d_1} = 0$, the bounds given by Theorem 1.12 do not improve (and is the same as) Deligne's bounds, Theorem 1.8. By contrast, Theorems 2.2 and 2.3 give the following better lower bounds.

- Before middle dimension, for $0 \leq m \leq e - 3$, we have:

$$\text{colevel of } H_c^{e^2-(e+1)+m}(X') \geq \left\lceil e - 1 - \frac{e-1}{e}(e-m) \right\rceil = \left\lceil \frac{(e-1)m}{e} \right\rceil.$$

The Esnault–Katz colevel lower bound in this case is 0.

- Beyond middle dimension, for $0 \leq j \leq e^2 - 4$, we have

$$\begin{aligned} \text{colevel of } H_c^{e^2-4+j}(X') &\geq j + \max \left\{ 0, \left\lceil \frac{e^2 - 4e + 3 - j}{e} \right\rceil \right\} \\ &= j + \max \left\{ 0, e - 4 + \left\lceil \frac{3 - j}{e} \right\rceil \right\}. \end{aligned}$$

When $e \geq 5$, this latter value is $j + e - 3$ if $0 \leq j \leq 2$, $j + e - 4$ if $3 \leq j \leq e + 2$, $j + e - 5$ if $e + 3 \leq j \leq 2e + 2$ etc. By contrast, the colevel lower bound given by Theorem 1.12 is j .

Acknowledgment

As noted, Theorem 1.12 has also been independently obtained by Rai and Shuddhodan as a by-product of [RS23], using a similar Lefschetz approach.

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DAQING WAN

CENTER FOR DISCRETE MATHEMATICS, CHONGQING UNIVERSITY, CHONGQING 401331 CHINA.

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF CALIFORNIA, IRVINE, CA 92697-3875, USA.

Email Address: `dwan@math.uci.edu`

DINGXIN ZHANG

YMSC, TSINGHUA UNIVERSITY, BEIJING 100086, CHINA.

Email Address: `dingxin@tsinghua.edu.cn`